

Education

Carnegie Mellon University

08/2022 – 06/2024 • Pittsburgh, PA

Master of Science in Robotic Systems Development • GPA: 3.92/4.0

- **Key Courses:** Learning for 3D Vision, Computer Vision, Visual Learning and Recognition, Robot Autonomy, AI Venture Studio

Manipal Institute of Technology

06/2017 – 05/2021 • Manipal, India

B.Tech Mechatronics (Minor in Robotics) • GPA: 8.78/10

- **Key Courses:** Machine Vision and Image Processing, AI, Intro to ML, Robot Kinematics, Robot Dynamics, Linear Algebra

Experience

Robotics Researcher

05/2023 – Present • Santa Clara, CA

Nissan Advanced Technology Center (Internship: May-Aug 2023, Full-time from Aug 2024)

- Engineered graph-based transformer pipeline for multi-object tracking, achieving 0.71 AMOTA with real-time inference on vehicle
- Developed Sparse4Dv3 LiDAR-Camera fusion tracker on nuScenes; hit 0.650 AMOTA, +0.17 over camera-only baseline
- Re-architected data pipeline to be detector-agnostic, enabling the integration of BEVFusion detections on the nuScenes benchmark
- Optimized 2b-param ViT for simulation; reached 9.2% MFU, 29x speedup by resolving silent upcasting and bottlenecks via profiling
- Built remote driving system's online pose estimation module using C++ and ROS achieving 50cm lateral and 0.5 deg yaw accuracy
- Pioneered first end-to-end LiDAR+Camera tracker (raw sensors to tracks, no separate detector); competitive with two-stage SOTA
- Engineered RANSAC-based ground plane detection with CUDA acceleration, achieving 10Hz real-time processing of point cloud
- Fused LiDAR data with on-board odometry using lag compensated EKF, resulting in 43% improvement in vehicle pose estimate
- Designed Hungarian track-matching architecture to identify ego vehicle from LiDAR tracked objects with 96% accuracy in real-time

Machine Learning Satellite Researcher

08/2023 - 08/2024 • Pittsburgh, PA

Robotics Exploration Lab, CMU

- Led a team of 5 to design a 2-stage ML approach for region classification and landmark detection to run on low-power edge devices
- Implemented an SPP classifier in PyTorch for vision-based region classification to run on NVIDIA Jetson Nano, achieving 94% mAP
- Augmented Landsat-8/9 data with conditional GANs (cGANs), simulating cloud cover and diurnal variations for dataset diversity
- Fine-tuned 18 YOLOv8n models as lightweight landmark detectors, reducing localization error to under 50 km at 700 km altitude

Computer Vision and Deep Learning Intern

01/2021 – 05/2021 • Hyderabad, India

RapidAI Vision, Defence Research and Development Organisation (DRDO)

- Architected vision-based precision landing system for a swarm of UAVs using fiducial marker landing pads from a height of 9 meters
- Conducted SIL, HIL tests with PX4, MAVROS and Gazebo, achieving 4% landing pad width error using embedded marker protocol
- Deployed custom ORBSLAM2 variant to reconstruct site with 90% accuracy using monocular camera and localized UAV in real-time

Projects

Open Source: Reviewer and contributor for Hugging Face LeRobot; merged PRs to JAX, PyTorch-RL, GTSAM, Transformers, and Rerun

- **lerobot-doctor** (42 ★) — dataset-quality diagnostics for LeRobot v3 — schema, statistics, integrity checks, and repair hints
- **vla-lite** — fixed a silent bug that left OpenVLA-7B blind on modern transformers; adds 4-bit inference on an 8 GB GPU
- **gtsam-splatfactors** (31 ★) — Gaussian-Splatting SLAM in GTSAM/iSAM2; featured on the official GTSAM blog
- **vggt-factor-refinement** (51 ★) — COLMAP-free SfM that refines VGGT 3D reconstruction with GTSAM factor graphs
- **recon3d** (27 ★) — one-command 3D reconstruction from a single handheld video — no COLMAP or manual camera calibration
- **lerobot-bench** — statistically rigorous A/B benchmarking of robot-learning policies

Zero-Shot Semantic Mapping for Mobile Manipulation

04/2026

Grand Prize Winner — RoboHacks @ Y Combinator (120 builders)

- Built spatial memory layer fusing Gemma VLM embeddings with SLAM pose graph for zero-shot object retrieval in novel scenes
- Trained ACT policy on Innate OS for 6-DoF pick-and-place; integrated the semantic map for language-conditioned task dispatch
- Won the Grand Prize and Best Use of Data; built and shipped the full perception-to-action stack on a MARS robot in under 2 days

Vision-Guided Robotic Cable Insertion

2026

Intrinsic (Alphabet) AI for Industry Challenge — Team HexDex, ranked 18th

- Trained and deployed a vision-guided cable-insertion policy on a physical robot cell — a task needing sub-5 mm placement accuracy
- Cut target pose error 7x (26–99 mm to 3.6 mm) by inferring it from a nearby well-localized part via known geometry
- Hardened the pipeline (startup, idempotency, and indexing bugs) and built offline evaluation tooling with ground-truth metrics

Skills

Programming Languages: Python, C++, C, CUDA, Java, JavaScript, MATLAB, Simulink, Bash

Frameworks and Libraries: PyTorch, JAX, CUDA, Hugging Face Transformers, LeRobot, TensorFlow, OpenCV, ROS

Tools and Platforms: Docker, Kubernetes, Gazebo, AirSim, Git, GitLab, AWS, GCP, Azure, Apache Kafka